

WEEK- 03

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Section:1[Practical]

PROGRAM-01

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int num, i;
    long long fact = 1;

    if (argc != 2) {
        printf("Usage: ./factorial <number>\n");
        return 1;
    }

    num = atoi(argv[1]);

    for (i = 1; i <= num; i++)
        fact *= i;

    printf("\n--- Child Process (exec) ---\n");
    printf("Factorial of %d = %lld\n", num, fact);

    return 0;
}
```

Output:

```
vaishnavi@Admin:~$ cd program1
vaishnavi@Admin:~/program1$ nano program1.c
vaishnavi@Admin:~/program1$ gcc program1.c -o factorial
vaishnavi@Admin:~/program1$ ./factorial 5

--- Child Process (exec) ---
Factorial of 5 = 120
vaishnavi@Admin:~/program1$ ./factorial 7

--- Child Process (exec) ---
Factorial of 7 = 5040
vaishnavi@Admin:~/program1$ ls
factorial  program1.c
vaishnavi@Admin:~/program1$ |
```

PROGRAM-02

```
#include <stdio.h>
#include <stdlib.h>

int main(int argc, char *argv[]) {
    int num, i;
    long long fact = 1;

    if (argc != 2) {
        printf("Usage: ./factorial <number>\n");
        return 1;
    }

    num = atoi(argv[1]);

    for (i = 1; i <= num; i++)
        fact *= i;

    printf("\n--- Child Process (exec) ---\n");
    printf("Factorial of %d = %lld\n", num, fact);

    return 0;
}
```

```

        exit(1);
    }

    if (pid == 0) {
        // Child Process
        sprintf(str, "%d", num);

        execl("./factorial", "factorial", str, NULL);

        // Executes only if exec fails
        perror("exec failed");
    }
    else {
        // Parent Process
        wait(NULL);

        printf("\n--- Parent Process ---\n");
        printf("Square of %d = %d\n", num, num * num);
    }

    return 0;
}

```

```

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>

int main() {
    int num;
    pid_t pid;
    char str[20];

    printf("Enter a number: ");
    scanf("%d", &num);

    pid = fork();

    if (pid < 0) {
        printf("Fork Failed\n");
        exit(1);
    }

    if (pid == 0) {
        // Child Process
        sprintf(str, "%d", num);

        execl("./factorial", "factorial", str, NULL);

        // Executes only if exec fails
        perror("exec failed");
    }
}

```

Output:

```
vaishnavi@Admin:~$ mkdir program2
vaishnavi@Admin:~$ cd program2
vaishnavi@Admin:~/program2$ nano program1.c
vaishnavi@Admin:~/program2$ gcc program1.c -o factorial
vaishnavi@Admin:~/program2$ nano program2.c
vaishnavi@Admin:~/program2$ gcc program2.c -o program2
vaishnavi@Admin:~/program2$ ./program2
Enter a number: 6

--- Child Process (exec) ---
Factorial of 6 = 720

--- Parent Process ---
Square of 6 = 36
vaishnavi@Admin:~/program2$ ls
factorial  program1.c  program2  program2.c
vaishnavi@Admin:~/program2$ |
```